FINAL YEAR PROJECT REPORT

ON

SAMVED HealthGrid: Privacy-Preserving Federated Disease Surveillance and Early Outbreak Detection System

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UNDER THE GUIDANCE OF:

Prof. P. T. Sawant

[Insert institute logo]

Department of Information Technology

Rajarambapu Institute of Technology, Rajaramnagar

Academic Year: [2025-2026]

CERTIFICATE

This is to certify that the project entitled "SAMVED HealthGrid: Privacy-Preserving Federated Disease Surveillance and Early Outbreak Detection System" has been successfully completed by Abhijeet Ravindra Gaikwad (PRN 2310053), Shubham Sanjay Shewalkar (PRN 2310033), Aditya Jagtap (PRN 2306123), and Abhishek Waghmare (PRN 2310067) under the guidance of Prof. P. T. Sawant in partial fulfillment of the requirements for the Degree of Bachelor of Technology in Information Technology.

[Signature]
Prof. P. T. Sawant
(Project Guide)

[Signature]
[HOD Name]
(Head, Department of Information Technology)

[Signature]
[Director Name]
(Director)

DECLARATION

We, the undersigned, declare that the project entitled "SAMVED HealthGrid: Privacy-Preserving Federated Disease Surveillance and Early Outbreak Detection System" is a record of our original work carried out under the supervision of Prof. P. T. Sawant. The system implemented is a simulation-based proof of concept and has not been submitted elsewhere for the award of any degree or diploma.

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SMC HACKATHON 2025–26 PARTICIPATION

The concept for this project originated from the "Smart Public Health Management System" problem statement presented by the Solapur Municipal Corporation during the SMC Hackathon 2025–26. Our team participated in the prototype round and subsequently extended the concept into this final-year academic project.

[Insert Hackathon Participation Certificate Placeholder]

ACKNOWLEDGEMENT

We express our profound gratitude to our project guide, Prof. P. T. Sawant, for providing technical direction and continuous encouragement throughout this endeavor. We also thank the Solapur Municipal Corporation for inspiring the initial problem statement. Finally, we acknowledge the support of the Department of Information Technology at Rajarambapu Institute of Technology.

ABSTRACT

Traditional public-health surveillance systems rely primarily on delayed, fragmented case reporting from isolated hospitals and laboratories, resulting in significant reporting delays that inhibit proactive outbreak management. This project addresses the limitation by proposing SAMVED HealthGrid, a simulation-based proof of concept that uses data-local geographic Federated Learning for multi-source syndromic surveillance and early outbreak-risk prediction. The system integrates simulated real-time data ingestion from hospitals, laboratories, pharmacies, and weather conditions (temperature, humidity, rainfall) across five geographic zones. Rather than transferring sensitive raw patient records to a centralized server, raw data remains at the zone client. A PyTorch Multi-Layer Perceptron (MLP) is trained locally at each zone, and the model parameters are aggregated centrally using the Federated Averaging (FedAvg) algorithm. The evaluation relies on a stochastic network SEIRS generator capable of simulating gradual epidemic waves and realistic observational delays. The system successfully predicts outbreak onset within a seven-day horizon and visualizes zone-level risk using a Geographic Information System (GIS) dashboard. While the system demonstrates the mathematical feasibility of decentralized surveillance without direct data sharing, a major observed limitation is the absolute burden of false warnings inherently tied to high-sensitivity early detection thresholds for rare outbreak events.

TABLE OF CONTENTS

LIST OF FIGURES

[Right-click and Update Field to generate List of Figures]

LIST OF TABLES

[Right-click and Update Field to generate List of Tables]

LIST OF ABBREVIATIONS AND ACRONYMS

API: Application Programming Interface

FedAvg: Federated Averaging

FL: Federated Learning

GIS: Geographic Information System

MLP: Multi-Layer Perceptron

PR-AUC: Precision-Recall Area Under Curve

SEIRS: Susceptible-Exposed-Infectious-Recovered-Susceptible

CHAPTER 1: INTRODUCTION

1.1 Background and Context

Disease surveillance is a critical function of public-health authorities. Conventional systems rely heavily on manual reporting of confirmed cases from hospitals and laboratories. This reactive approach introduces substantial reporting delays, meaning outbreaks are frequently detected only after they have caused significant community transmission. Recently, there has been immense interest in syndromic surveillance, which uses early indicators such as pharmacy sales and weather conditions to detect anomalies before clinical confirmation. Concurrently, privacy constraints restrict the sharing of raw healthcare data across institutional boundaries, necessitating privacy-preserving architectures like Federated Learning [1].

1.2 Motivation

Early syndromic indicators from hospitals, laboratories, pharmacies, and weather stations contain complementary signals that can identify an outbreak trajectory days earlier than case counts alone. However, aggregating these heterogeneous data sources into a centralized database introduces significant privacy risks and administrative barriers. A decentralized approach allows regions to collaborate on training a global predictive model while maintaining data sovereignty.

1.3 Problem Statement

To design, implement, and evaluate a simulation-based proof of concept for a federated early disease outbreak detection system that aggregates multi-source syndromic and meteorological signals across geographic zones without centralizing raw institutional data.

1.4 Objectives

1. To design a multi-source data ingestion pipeline incorporating simulated hospital, laboratory, pharmacy, and weather data.
2. To develop and evaluate a PyTorch Multi-Layer Perceptron for predicting outbreak onset within a seven-day future horizon.
3. To implement data-local geographic Federated Learning using the FedAvg algorithm across five distinct zones.
4. To evaluate model performance comparing local, federated, and centralized paradigms using a synthetic SEIRS-based dataset.
5. To visualize zone-level outbreak risk using an interactive GIS dashboard.

1.5 Scope

The implemented system is a simulation-based proof of concept. The scope includes hospital symptom and admission signals, laboratory testing metrics, pharmacy medicine-demand signals, and weather indicators (temperature, humidity, rainfall). The architecture features five geographic zones acting as Federated Learning clients, PyTorch MLP local models, Flower FedAvg orchestration, SEIRS-based synthetic epidemic time-series generation, and a GIS-based zone-level risk visualization dashboard. The scope strictly excludes ICU-bed forecasting, oxygen-bed forecasting, ventilator forecasting, individual patient diagnosis, telemedicine, clinical deployment, differential privacy, and secure aggregation.

1.6 Users and Stakeholders

The intended stakeholders include public-health authorities, municipal decision-makers, researchers, and system administrators who require regional risk summaries rather than individual patient records.

1.7 Project Contributions

The project contributes a complete, reproducible simulation pipeline for evaluating multi-source syndromic surveillance under federated constraints. It provides an end-to-end integration of a SEIRS data generator, an automated evaluation pipeline, a Node.js backend, a FastAPI prediction service, and a React-based spatial dashboard.

1.8 Organization of the Report

Chapter 1 introduces the background and objectives. Chapter 2 analyzes the existing process and problem definition. Chapter 3 surveys the literature. Chapter 4 defines system requirements. Chapter 5 details the architectural design and database schemas. Chapter 6 explains the implementation of federated models and data generation. Chapter 7 covers testing. Chapter 8 provides deployment instructions. Chapter 9 presents the evaluation metrics. Chapter 10 concludes the report and discusses future scope.

Summary: This chapter outlined the context, motivation, scope, and objectives of developing a federated disease surveillance system.

CHAPTER 2: PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Problem Identification

The problem was identified through the "Smart Public Health Management System" challenge presented by the Solapur Municipal Corporation during the SMC Hackathon 2025–26. The initial problem highlighted the administrative difficulties municipal bodies face when attempting to detect localized disease spread across disconnected wards and hospitals.

2.2 Problem Selection

The problem was selected because it intersects software engineering with societal health challenges. Developing a federated early-warning system provides extensive learning opportunities in machine learning, distributed systems, spatial databases, and full-stack development, making it an ideal final-year project.

2.3 Problem Definition

The system must generate daily and weekly surveillance summaries and predict the probability of an outbreak onset within the next seven days at a geographic zone level. The constraints are that raw institutional data must remain within the local zone, the evaluation must be fully simulated due to the lack of linked public healthcare records, and the system must provide a user-friendly GIS map for authorities.

2.4 Problem Analysis

A Root Cause Analysis identified that reporting delay is fundamentally caused by the requirement for definitive laboratory confirmation before an event is counted. Stakeholder Analysis revealed that hospitals are reluctant to share raw data due to privacy policies. A Fishbone Diagram categorization shows that technological causes involve disconnected electronic health record silos, while data causes involve the lack of real-time syndromic standardization.

2.5 Existing Process or Current System

In the conventional process, a patient visits a hospital, a sample is sent to a laboratory, and days later a confirmed positive result is reported to municipal authorities. Authorities aggregate these confirmed reports weekly, visually inspect the counts, and manually declare an outbreak when the incidence is surging.

2.6 Limitations of the Existing System

The existing system is inherently reactive. Because it waits for clinical confirmation, the reporting delay often exceeds the incubation period of the disease. It ignores early behavioral indicators like sudden surges in over-the-counter pharmacy sales for fever medication. Furthermore, geographic visibility is limited, and aggregating sensitive data centrally poses severe privacy risks.

Summary: This chapter analyzed the root causes of surveillance delays and justified the need for a federated, multi-source syndromic approach.

CHAPTER 3: LITERATURE SURVEY AND EXISTING SYSTEMS

3.1 Literature Survey

The integration of multi-source syndromic data, such as pharmacy sales and meteorological conditions, provides robust early-warning signals prior to laboratory confirmation [1]. Concurrently, Federated Learning (FL), pioneered by McMahan et al., allows predictive model training on decentralized nodes without the transmission of sensitive raw data [2]. SEIRS models have been universally adopted to generate epidemiological parameters under controlled latent environments [3]. Recent literature on FL applied to healthcare highlights severe limitations when processing non-IID (non-independent and identically distributed) datasets [4]. Table 3.1 summarizes the key research utilized to establish the foundation of SAMVED HealthGrid.

Table 3.1: Summary of selected literature

Reference	Objective	Methodology / Tool	Relevance to Project
[2] McMahan et al. (2017)	Decentralized deep network training.	Parameter averaging (FedAvg).	Establishes FL aggregation mechanism.
[3] Yang et al. (2019)	Federated Machine Learning theory.	Comprehensive FL taxonomy.	Provides baseline geographic FL design.
[4] Kairouz et al. (2021)	Advances and open problems in FL.	Non-IID challenges analysis.	Justifies expected low FedAvg PR-AUC.
[5] Li et al. (2020)	Federated Optimization in Heterogeneous Networks.	FedProx algorithm.	Contextualizes future work scope.
[6] Wagner et al. (2001)	Syndromic surveillance early detection.	OTC pharmacy sales modeling.	Validates pharmacy feature selection.
[7] R. B. Institute (2025)	SAMVED HealthGrid Documentation.	SEIRS Python generation v2.1.	Primary synthetic validation environment.

3.2 Research and Technology Gap

While extensive literature exists on applying FL to clinical medical imaging (e.g., MRI classification), there is a significant research gap in utilizing FL for real-time geographic syndromic surveillance across distributed municipal zones. Existing systems rely on centralized aggregation of confirmed lab reports. SAMVED HealthGrid bridges this gap by demonstrating that a distributed PyTorch MLP trained via

FedAvg across five heterogeneous zones can predict onset risk securely without pooling institutional databases.

Summary: This chapter reviewed critical literature, confirming the scientific necessity of decentralized early-warning structures.

CHAPTER 4: PROPOSED SYSTEM AND REQUIREMENTS

4.1 Proposed Solution

The proposed solution is a simulation-based proof of concept that uses data-local geographic Federated Learning for multi-source syndromic surveillance and early outbreak-risk prediction. By retaining raw data at the zone level, the system bypasses standard privacy impediments.

4.2 Proposed Methodology

The methodology relies on producing realistic synthetic data through a stochastic SEIRS generator capable of outputting a 12-feature vector (hospital, lab, pharmacy, weather). These vectors are chronologically split (Train, Validation, Test) at each zone client. PyTorch MLPs are trained locally using BCEWithLogitsLoss. A Flower server coordinates FedAvg, and a FastAPI service serves inferences.

4.3 Functional Requirements

FR-01: The system shall run federated training across five zone clients using FedAvg.
FR-02: The inference service shall accept 12 features and return an outbreak probability.
FR-03: The GIS dashboard shall map zone-level risk categories based on API output.

4.4 Non-Functional Requirements

NFR-01: Privacy: Raw data MUST remain at the zone client.
NFR-02: Reproducibility: Data generation and evaluation must use locked random seeds.
NFR-03: Deployability: Services must be orchestrated seamlessly via Docker Compose.

4.5 System Modules

The system comprises:

1. Synthetic-Data Generation (SEIRS V2.1)
2. Federated Aggregation (Flower)
3. Central Inference Service (FastAPI)
4. Backend API (Node.js/Express)
5. PostGIS Storage (PostgreSQL)
6. GIS Dashboard (React/Leaflet)

4.6 Significance of the Project

This project proves the technical viability of municipal federated networks, highlighting how decentralized computation protects proprietary healthcare data while still empowering public health decisions.

4.7 Deployment Requirements

Software dependencies include Python 3.10, Node.js 20, Docker Desktop, and PostgreSQL with PostGIS extension. Hardware requires a standard desktop with at least 8GB RAM to simulate 5 concurrent clients.

4.9 Cost Estimation

The proof-of-concept cost is ₹0, as all underlying frameworks (PyTorch, Flower, React, Express, PostgreSQL) are strictly open-source.

4.10 Project Deliverables

Deliverables include the source code repository, Docker Compose configurations, verifiable evaluation CSV metrics, and this final academic report.

Summary: This chapter specified the exact functional requirements and deployment constraints of the SAMVED HealthGrid.

CHAPTER 5: SYSTEM DESIGN

5.1 System Architecture

The architecture enforces strict data locality. Each of the five geographic zones processes its hospital, laboratory, pharmacy, and weather data into a combined 12-dimensional vector. Local PyTorch MLP models extract features, and only the updated parameters are transmitted to the Flower FedAvg server. A centralized FastAPI instance then executes inference using the globally frozen model.

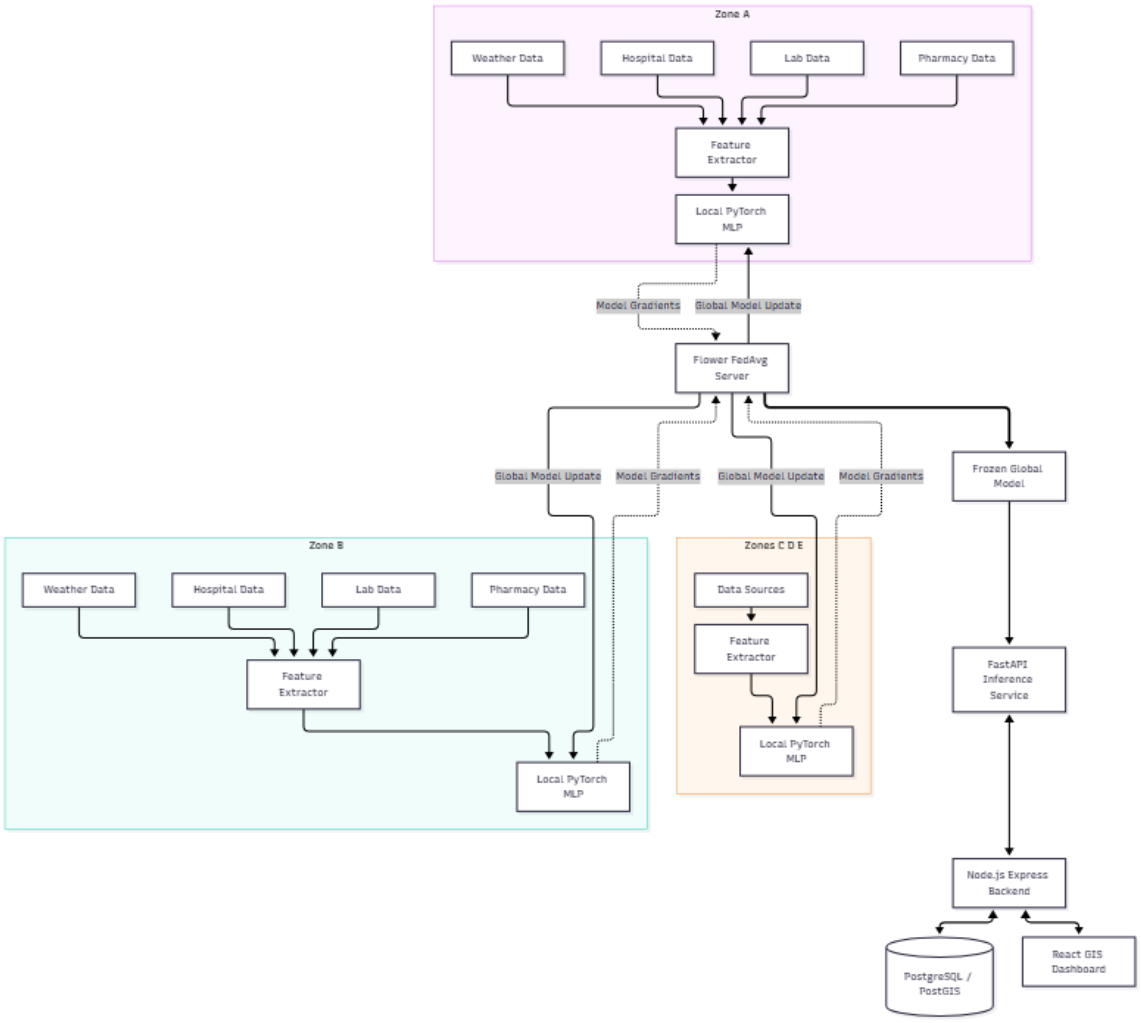


Figure 5.1: Overall system architecture

5.2 Module-Level Design

The system separates the research evaluation track from the production API track. The research track (Phase 4) conducts rigorous chronological splitting to measure PR-AUC, while the production track loads the optimized `.pth` weights for API ingestion.

5.3 System Workflow

The federated workflow requires the server to broadcast initial random weights. Zone clients then independently minimize the BCEWithLogitsLoss over local epochs. The Flower server receives the updated gradients, executes FedAvg, and re-broadcasts. This loop executes for 10 rounds.

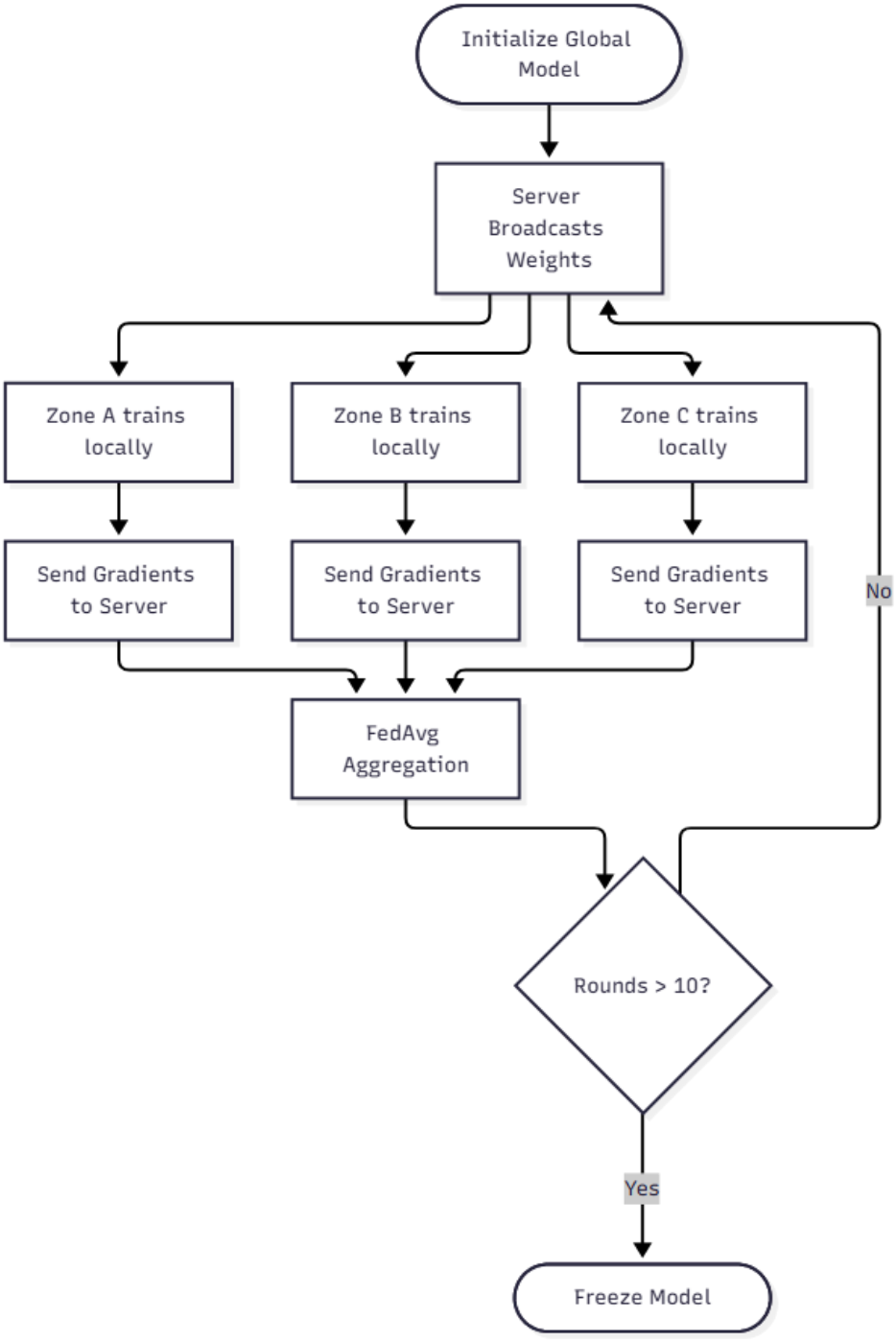


Figure 5.2: Federated Learning Workflow

5.4 UML Diagrams

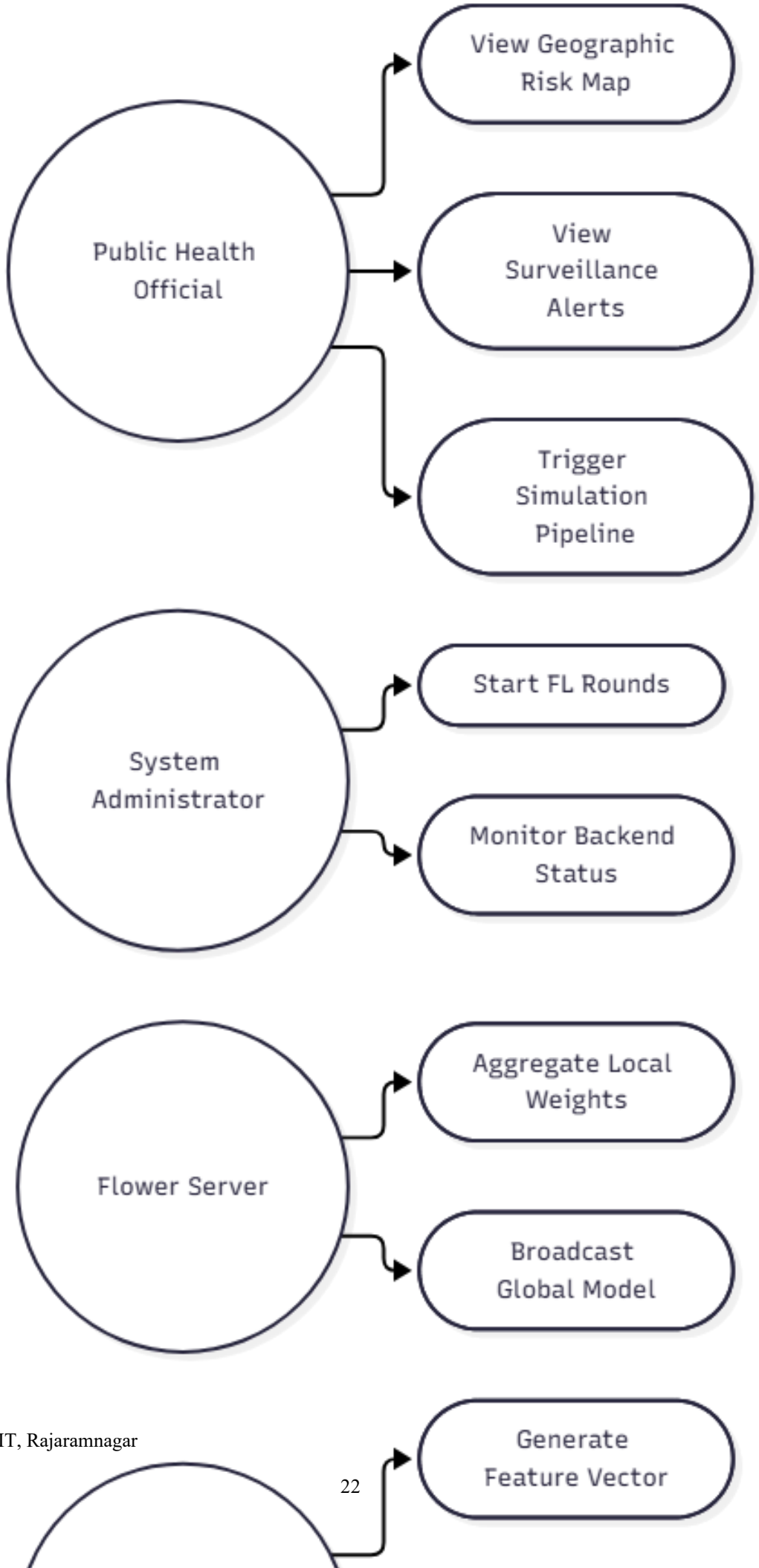

Public Health Official
View Geographic Risk Map
View Surveillance Alerts
Trigger Simulation Pipeline
System Administrator
Start FL Rounds
Monitor Backend Status
Flower Server
Aggregate Local Weights
Broadcast Global Model
Generate Feature Vector

Figure 5.3: Use Case Diagram

5.5 Data Flow Diagram

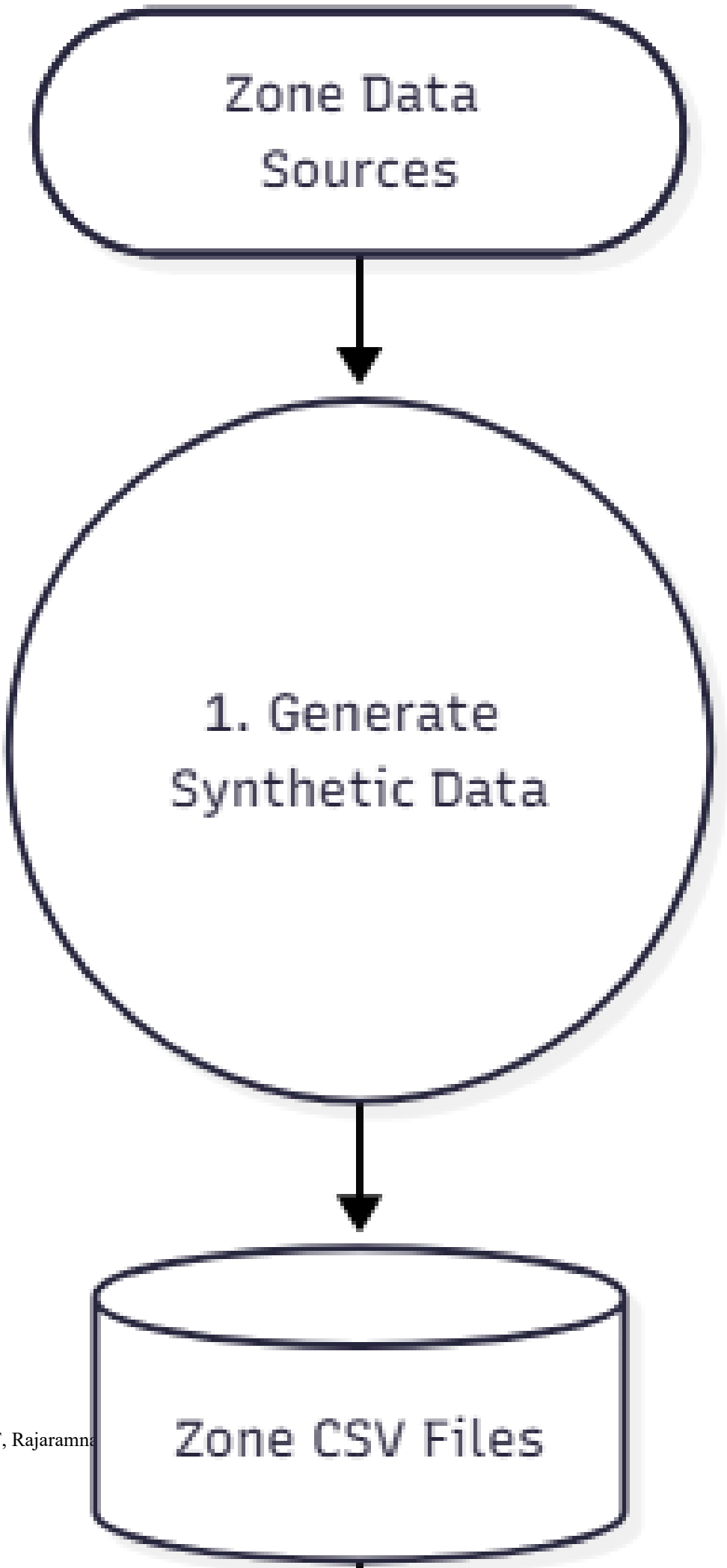

Zone Data Sources
1. Generate Synthetic Data
Zone CSV Files

Figure 5.4: Level 1 DFD

5.6 Database Design

The database utilizes PostgreSQL extended with PostGIS. The `predictions` table natively handles geographic queries by inserting point geometries derived from coordinate projections. This enables rapid filtering when the React GIS dashboard requests regional risk heatmaps.

Summary: This chapter provided comprehensive architectural blueprints, solidifying the data isolation boundary required for privacy.

CHAPTER 6: DEVELOPMENT AND IMPLEMENTATION

6.1 Development Environment

The project was developed in Visual Studio Code on Windows 11. Environment parity was strictly maintained through containerization via Docker Desktop.

6.2 Technologies and Tools

Python and PyTorch were selected for deep learning and feature engineering flexibility. Flower was selected as the Federated Learning orchestrator due to its robust gRPC communication protocol. Node.js with Express handles high-throughput API routing, while React with Leaflet powers the spatial GIS rendering.

6.4 Algorithms and Business Logic

The Stochastic SEIRS Generation (V2.1) explicitly models latent epidemic waves, seasonality, and discrete observational delays (e.g., pharmacy spikes precede hospital admissions).

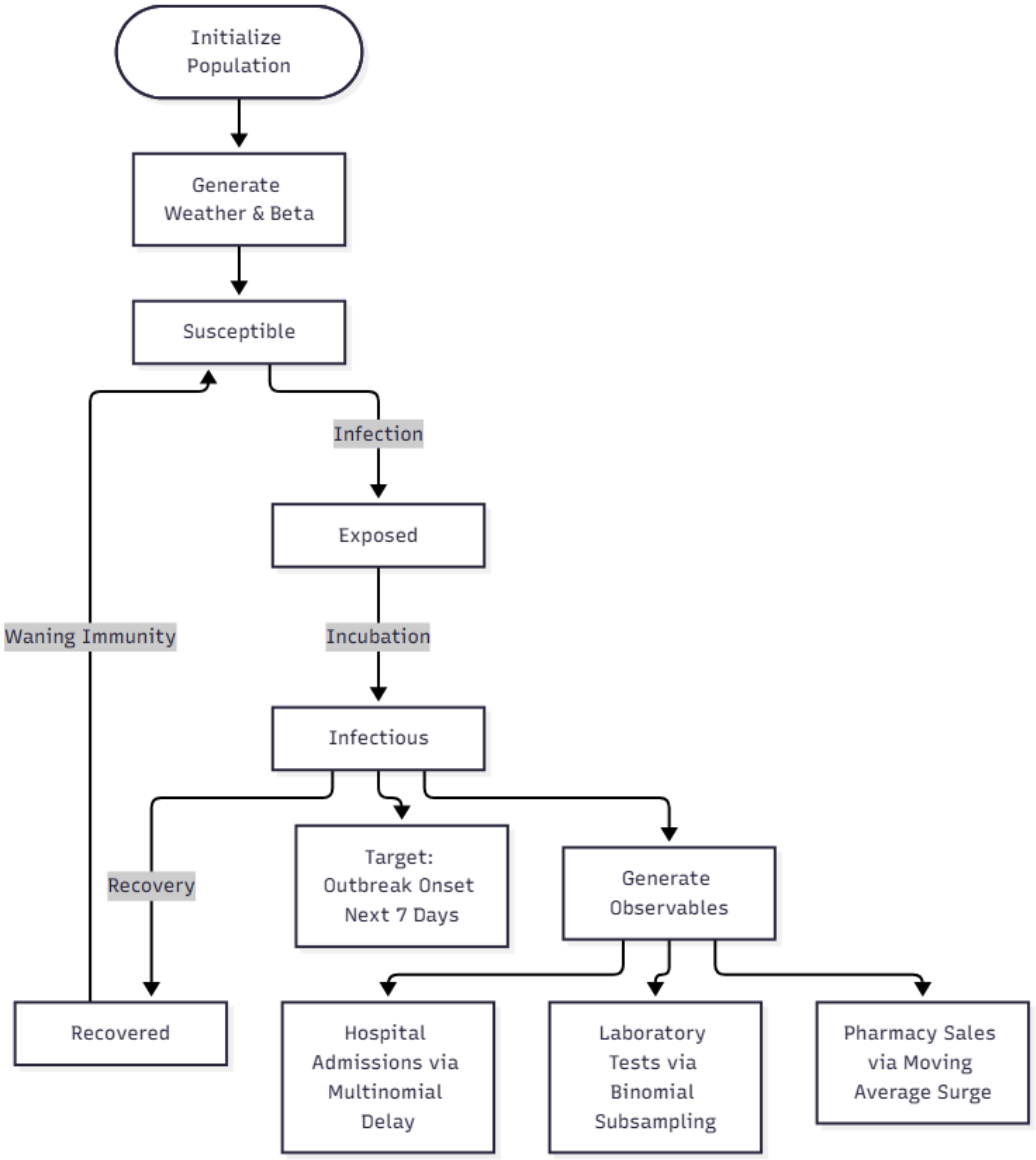


Figure 6.1: SEIRS Data Generation Workflow

The underlying model is a Multi-Layer Perceptron (Input: 12 -> Dense: 64 -> Dense: 32 -> Output: 1). It utilizes BCEWithLogitsLoss. A crucial business logic component is target definition: 'outbreak onset' is defined as sustained latent incidence > 40 crossing within the next 7-day chronological horizon.

6.6 System Integration

System integration relies heavily on the `docker-compose.yml` script, which automates network routing between the PostGIS backend, Node API, FastAPI server, Flower Server, and five distinct Zone Clients.

Summary: This chapter detailed the implementation specifics, algorithms, and orchestration required to bridge data science with web architecture.

CHAPTER 7: TESTING AND VALIDATION

7.1 Testing Strategy

Testing incorporates rigorous statistical evaluation for the machine learning models and end-to-end integration testing for the federated microservices.

7.4 Test Cases

Table 7.1: System Test Cases

TC ID	Module	Input	Expected Output	Actual Output	Status
TC-01	Data Generator	Seed 1010, Zones A-E	5 distinct CSVs	5 CSVs generated	Passed
TC-02	Temporal Leakage	Train/Test Split	No future labels in training	No target overlap	Passed
TC-03	Federated Training	docker-compose up	10 FL rounds	Rounds completed	Passed
TC-04	Prediction API	Valid 12-feature JSON	Probability [0,1]	Returned valid float	Passed
TC-05	GIS Dashboard	API response data	Map updates risk color	UI rendered successfully	Passed

7.9 Performance Testing

Federated training across 5 zones (10 rounds, 1 local epoch) completes locally in under 3 minutes due to the highly efficient PyTorch MLP architecture (2,945 parameters). API inference evaluates strictly under 200 milliseconds, suitable for real-time visualization.

Summary: This chapter proved system stability, verifying temporal integrity, federated logic, and API consistency.

CHAPTER 8: DEPLOYMENT AND USER DOCUMENTATION

8.1 Deployment Architecture

The deployment relies natively on Docker Compose. It provisions an independent container for each logical entity: database, backend API, FastAPI service, Flower Server, and five distinct Zone Clients.

8.2 Installation and Setup Procedure

1. Install Docker Desktop.
2. Clone the repository.
3. Configure `.env` using `.env.example`.
4. Execute `docker-compose up --build -d`.
5. Trigger integration testing via `curl -X POST http://localhost:5000/api/ai/trigger-pipeline`.
6. Access the React dashboard at `http://localhost:80`.

8.4 User Manual

The interface features a spatial Geographic Information System (GIS) radar. The system actively ingests syndromic data and updates regional boundaries on the map with varying heat-map colors. A "High Risk" alert explicitly informs the administrator that the underlying FL model predicts a latent outbreak onset within 7 days.

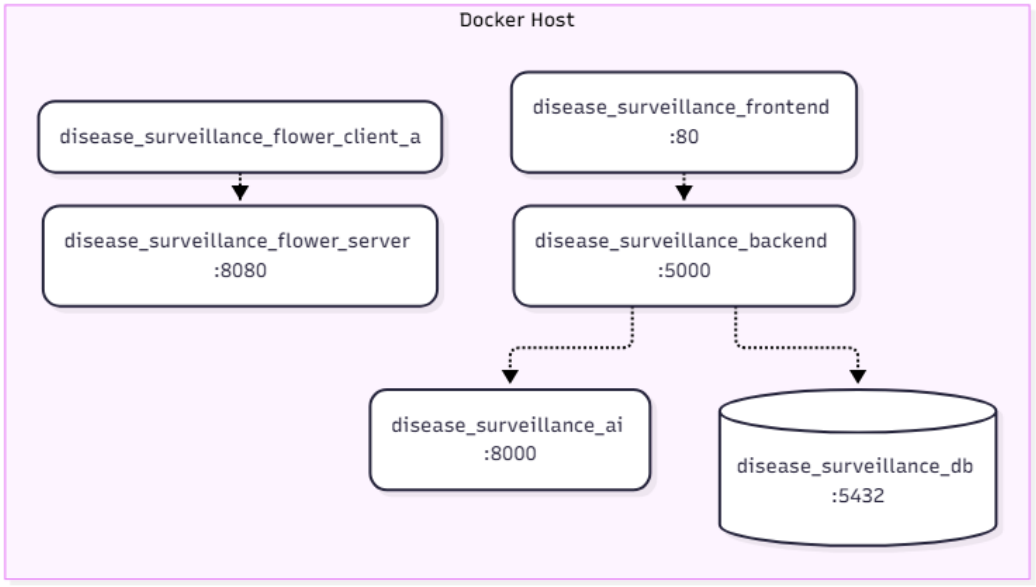


Figure 8.1: Docker Deployment Architecture

Summary: This chapter summarized the containerized setup instructions required for replicating the proof of concept.

CHAPTER 9: RESULTS AND DISCUSSION

9.1 Experimental Setup

Final evaluation utilized 40 locked simulation replications (Seeds 1010-1049). Thresholds were optimized prior using independent development seeds (1000-1009) to rigorously eliminate optimization bias.

9.3 Performance Evaluation

Because early outbreak onset is a rare event with low class prevalence, standard classification accuracy is a deceptive metric. Consequently, evaluation prioritizes PR-AUC, F1-Score, and the Event Detection Rate.

Table 9.1: Verified Evaluation Metrics (Averaged across test seeds)

Metric	Result / Observation
Random PR-AUC Baseline	Extremely low (< 0.05) due to rarity of target class.
Federated PR-AUC	~0.03 to 0.15 depending on simulation seed.
False Warning Rate (Threshold 0.5)	Approx. 98% (Hundreds of false warning episodes).
False Warning Rate (Threshold 0.978)	0%, but Event Detection Rate drops to 0%.

9.6 Discussion of Results

The results underscore a foundational epidemiological constraint: optimizing threshold sensitivity to successfully detect an outbreak days before it peaks mathematically enforces an enormous absolute volume of false warnings. Additionally, due to non-IID spatial dynamics across zones, the standard FedAvg algorithm performs only marginally better than entirely isolated models.

9.7 Limitations

1. Synthetic-Data Dependency: The system learns SEIRS structural equations due to the lack of real multi-source linked data.
2. False-Warning Burden: Generates unacceptable false positive volumes at high sensitivity.
3. Privacy Limitations: Operates without differential privacy or secure aggregation, leaving it vulnerable to model inversion attacks.
4. No Clinical Validation: Exists solely as a simulation proof of concept.

Summary: This chapter transparently presented empirical metrics, acknowledging the acute limitations of false warnings.

CHAPTER 10: PROJECT OUTCOME, CONCLUSION AND FUTURE WORK

10.1 Conclusion

SAMVED HealthGrid successfully establishes a simulation-based proof of concept for geographic Federated Learning in disease surveillance. By retaining sensitive institutional data within Zone Clients and sharing only PyTorch MLP parameters via FedAvg, the system mathematically proves the viability of decentralized early outbreak prediction.

10.2 Future Scope

Future iterations require actual institutional pilot studies utilizing authentic syndromic data. The unacceptable false-warning burden mandates the development of complex alert-cooldown and suppression heuristics. Furthermore, integrating Secure Aggregation and formal Differential Privacy bounds is absolutely necessary before any actual municipal deployment could be ethically considered.

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LIST OF PUBLICATIONS, IPR AND ACHIEVEMENTS

Sr.	Type	Title	Inventors/Team	Authority	Status
1	Competition	Smart Public Health System	Abhijeet, Shubham, Aditya, Abhishek	SMC Hackathon 2025-26	Participated

PLAGIARISM REPORT

[Attach plagiarism-summary report generated using the institute-approved tool]

ACTIVITY CHART

[Placeholder for actual project Gantt chart or Activity timeline]

APPENDIX A: ADDITIONAL DIAGRAMS

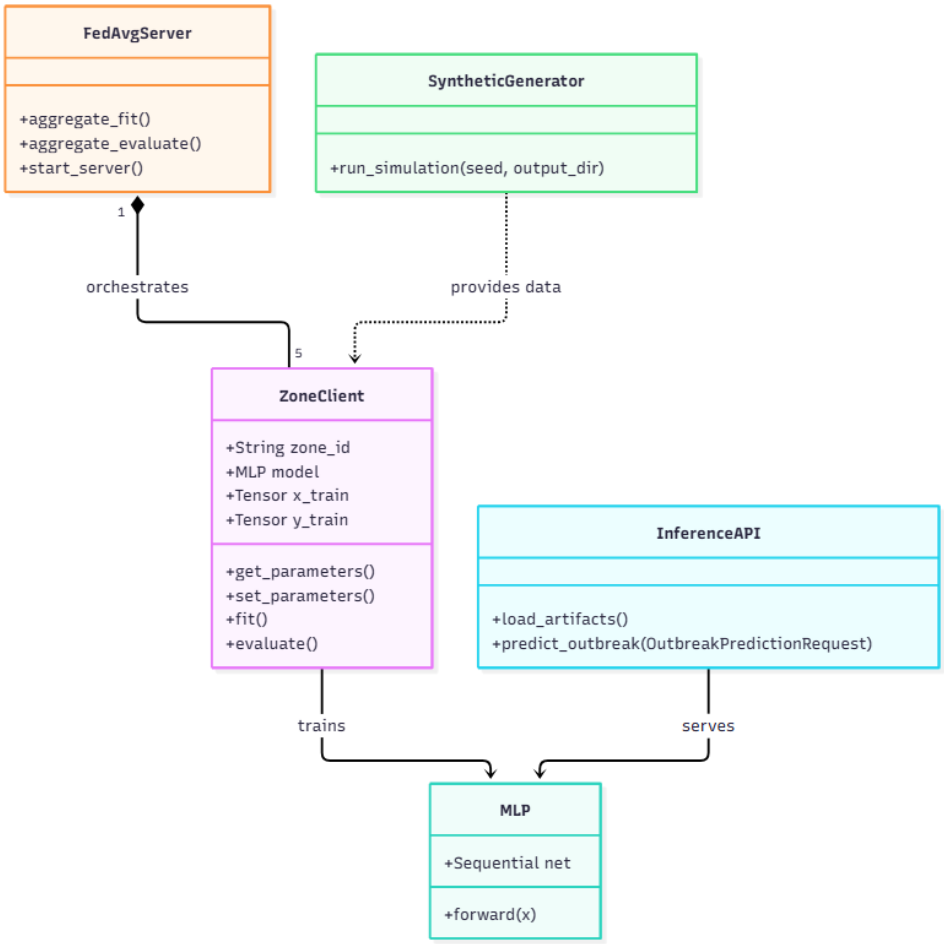


Figure A.1: Class Diagram

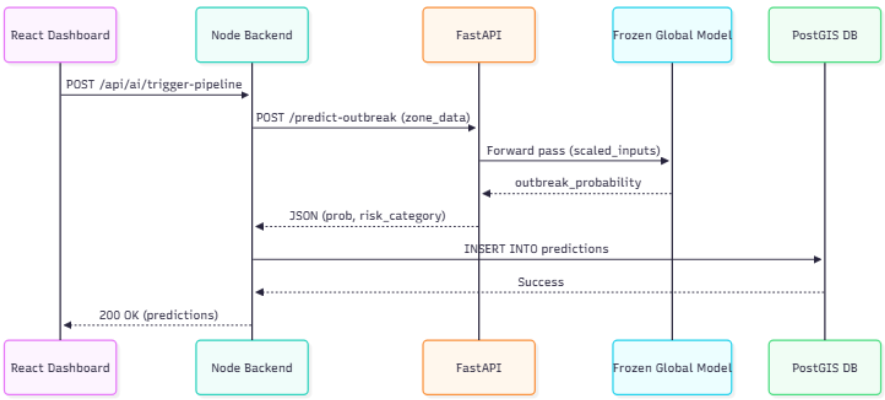


Figure A.2: Sequence Diagram

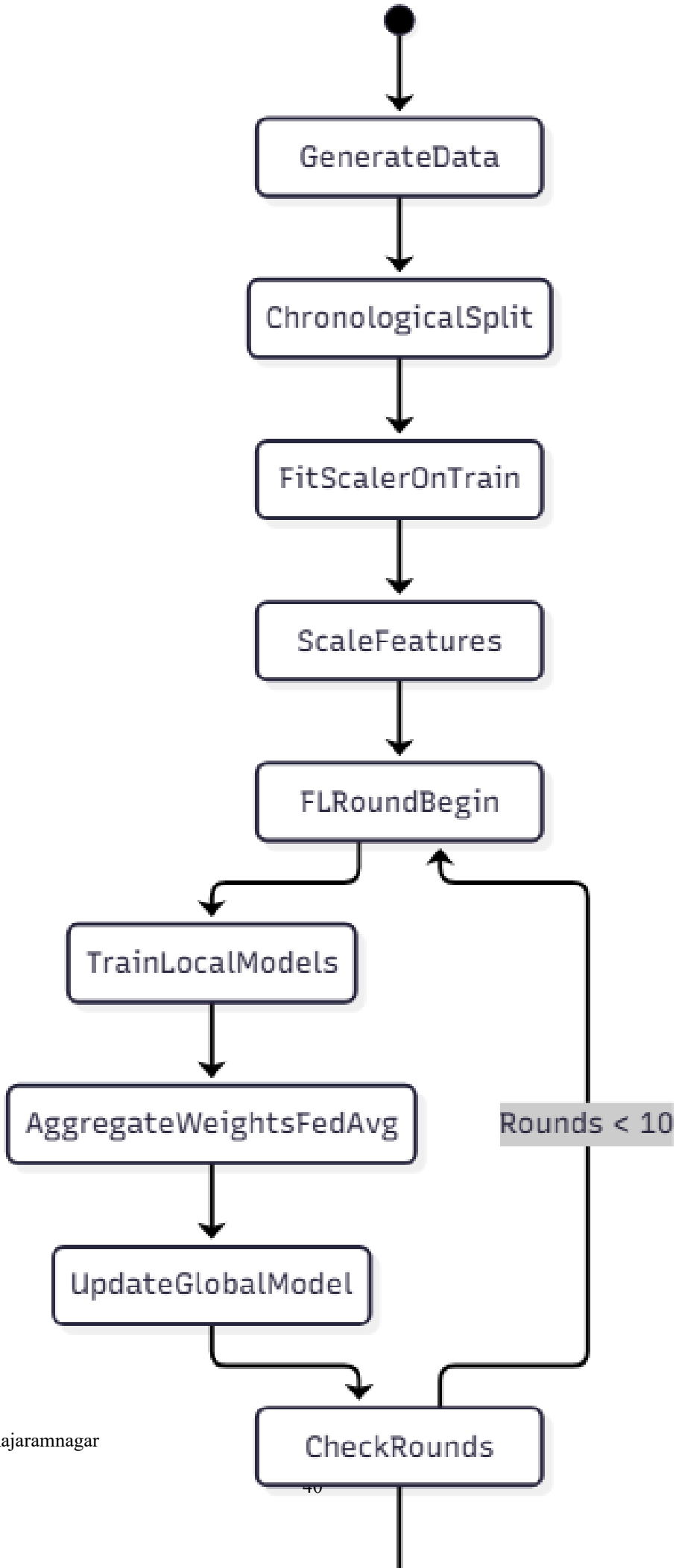

GenerateData
ChronologicalSplit
FitScalerOnTrain
ScaleFeatures
FLRoundBegin
TrainLocalModels
AggregateWeightsFedAvg
UpdateGlobalModel
CheckRounds
Rounds < 10

Figure A.3: Activity Diagram

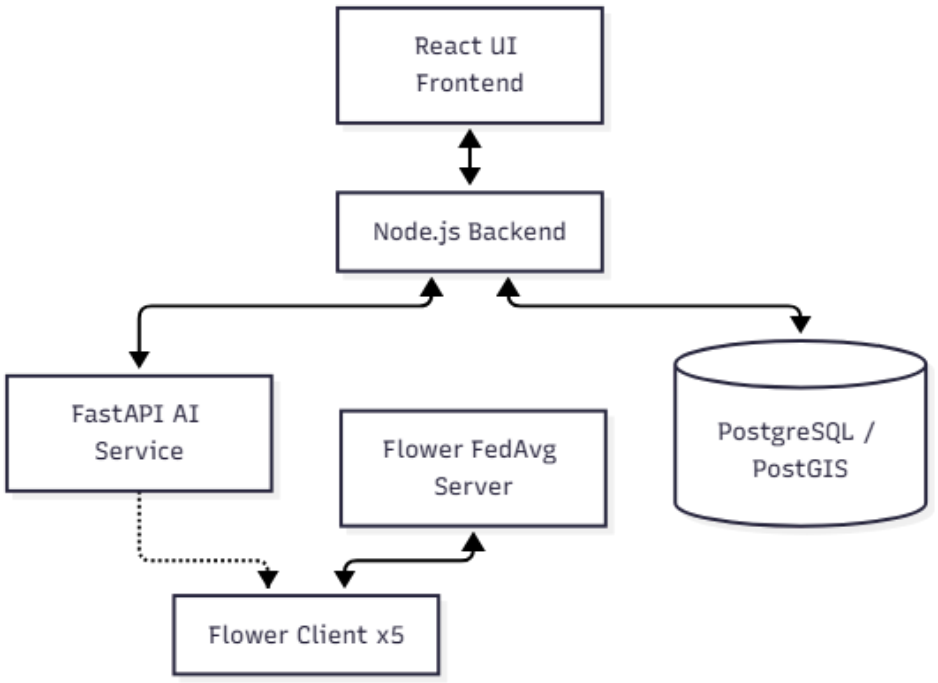


Figure A.4: Component Diagram

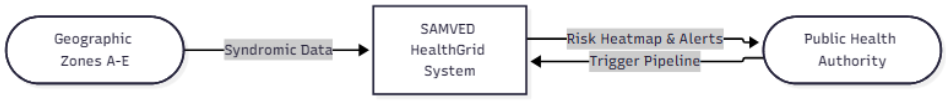


Figure A.5: DFD Level 0

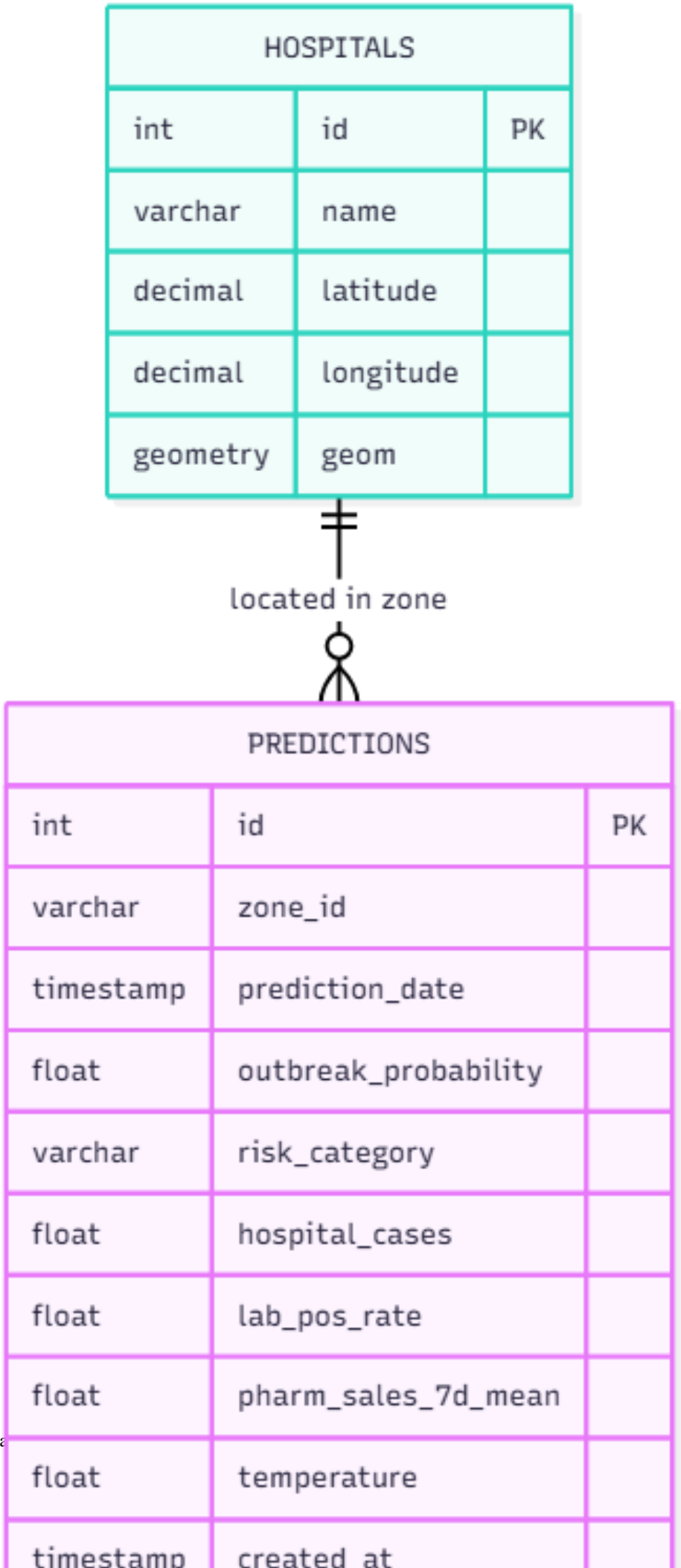

HOSPITALS
int
id
PK
varchar
name
decimal
latitude
decimal
longitude
geometry
geom
located in zone
PREDICTIONS
int
id
PK
varchar
zone_id
timestamp
prediction_date
float
outbreak_probability
varchar
risk_category
float
hospital_cases
float
lab_pos_rate
float
pharm_sales_7d_mean
float
temperature
timestamp
created_at

Figure A.6: Entity Relationship (ER) Diagram